

AKANSHA SUBEDI

STRUCTURAL ENGINEER

Melbourne, VIC (Australian Permanent Resident)

[Portfolio website](#)

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STATEMENT

Structural Engineer with 6+ years of experience in the analysis and design of steel and reinforced concrete structures across building and bridge projects. Recently relocated to Melbourne as an Australian Permanent Resident, with experience applying ACI, AISC, ASCE, Eurocode, Indian, Thai, and Nepali standards and working familiarity with Australian Standards (AS/NZS). Skilled in finite element analysis (FEA), structural design, performance-based design, ETABS, SAP2000, and Revit.

WORK EXPERIENCES

Structural Design Engineer (*Freelance – Thailand & Nepal*)

Jan 2024 – Present

- Performed structural analysis and design of numerous projects including steel framed residential/ resorts buildings, car parks, warehouses, a sports facility with truss roofing and RC framed commercial complexes & community center in Thailand and Nepal in accordance with ACI, AISC, ASCE, Thai (DPT 1311-50), IS 456, and NBC codes.
- Designed foundations, steel connections, base plates, and anchor bolts using engineering calculations, ensuring code compliance and practical constructability.
- Prepared structural calculations, reports, drawings and design documentation while coordinating with architects, clients and contractors.

Structural Engineer (*Computers and Structures Inc. (CSI) Bangkok*)

June 2023 – Dec 2023

- Performed verification testing of bridge analysis and design software (CSiBridge-ET), Structural detailing software (CSiDetail) and RC column design software (CSiCol) validating analysis, design, and detailing functionality against engineering standards.
- Performed manual calculations of concrete bridge components and verified software results.
- Designed PSC girders, bent caps, columns using AASHTO LRFD, Caltrans, and Eurocode.
- Developed algorithms and workflow documentation for analysis and design process.
- Prepared technical reports and collaborated remotely with international QA and dev teams.

Civil Engineer (Structural Design) (*TEAM Consultants Pvt. Ltd., Nepal*)

April 2017 – Oct 2019

- Performed linear static and dynamic analysis, calculations, and seismic design of more than 15 building structures
- Prepared structural designs, drawings, calculation reports, BOQs and cost estimates for low- to mid-rise residential and commercial buildings, a commercial complex with an RC core shear wall system, and a hospital extension with an expansion joint.
- Conducted site inspections, coordinated with architects, contractors, and local authorities, and resolved construction-related technical issues.
- Collaborated with architects, MEP engineers, geotechnical engineers, and clients throughout the design process to develop practical and coordinated structural solutions.

EDUCATION

Master of Engineering (Structural Engineering)

2020 – 2022

Asian Institute of Technology, Thailand

Bachelor of Civil Engineering

2011 – 2015

Tribhuvan University, Nepal

CORE COMPETENCIES

- Structural Analysis & Design
- Steel & Reinforced Concrete Design
- Seismic & Wind Engineering
- Performance-Based Design (PBD)
- Finite Element Analysis (FEA)
- Structural Assessment
- Calculation and Design Documentation

TECHNICAL SKILLS

- Structural Analysis & Design: **ETABS, SAP2000, SAFE, PERFORM-3D, CSiBridge, STAAD.Pro, VecTor FormWorks, SpaceGass, RAPT**
- Drafting & BIM: **AutoCAD, Revit**
- Programming & Analysis: **Python, MATLAB, Microsoft Excel**

RESEARCH EXPERIENCE

Thesis in Master in Structural Engineering

2021-2022

Topic: "The Effect of Performance-Based Wind Design on Seismic Design of a Tall Building in High Seismic and High Wind Zone"

Thesis Advisor: *Dr. Naveed Anwar*

- 93 Storied building: RCC framed core wall and BRB provided for lateral force resistance
- Studied the effect of considering non-linearity in wind design on seismic ductile design of the building through performance-based approach

Student Research Assistant (*Asian Institute of Technology*)

Jan 2021– April 2022

- Contributed to the development of the textbook "Strut and Tie Method" through literature review and technical writing.
- Conducted research on Advance Concrete Structures and Tall Buildings, and developed academic articles
- Conducted software testing and technical documentation for structural engineering software, contributing to the development of CSiDetail & CSi Designer.

Intern Engineer (*HELVETAS Swiss Inter-cooperation, Nepal*)

June 2016– March 2017

- Performed complete survey of water supply system from the water source to households and designed infrastructures such as intake, reservoirs, and pipelines

Intern Engineer (*Hydro-Consult Engineering Ltd., Nepal*)

April 2016– May 2016

- Supported structural assessments of intake structures, spillways, and powerhouse buildings
- Prepared technical drawings and FEM models under supervision of senior engineers

LICENSE AND CERTIFICATE

Engineers Australia - Migration Skills Assessment (Civil Engineer – Structural)

August 2024

Nepal Engineering Council - Registered Professional Engineer

Since June 2016

PUBLICATIONS

- ["Moment Curvature Curve: Unlocking the Hidden Treasures"](#)
- ["Basic Concepts for Considering Soil-Structure Interaction"](#)